Category

Equity Valuation: Concepts and Basic Tools

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THE GORDON GROWTH MODEL

Learning Outcomes

- calculate and interpret the intrinsic value of an equity security based on the Gordon (constant) growth dividend discount model or a two-stage dividend discount model, as appropriate
- identify characteristics of companies for which the constant growth or a multistage dividend discount model is appropriate
- explain advantages and disadvantages of each category of valuation model

A rather obvious problem when one is trying to implement Equation 1 for common equity is that it requires the analyst to estimate an infinite series of expected dividends. To simplify this process, analysts frequently make assumptions about how dividends will grow or change over time. The Gordon (constant) growth model (Gordon, 1962) is a simple and well-recognized DDM. The model assumes dividends grow indefinitely at a constant rate.

Because of its assumption of a constant growth rate, the Gordon growth model is particularly appropriate for valuing the equity of dividend-paying companies that are relatively insensitive to the business cycle and in a mature growth phase. Examples might include an electric utility serving a slowly growing area or a producer of a staple food product (e.g., bread). A history of increasing the dividend at a stable growth rate is another practical criterion if the analyst believes that pattern will hold in the future.

With a constant growth assumption, Equation 1 can be written as Equation 8, where g is the constant growth rate:

$$V_0 = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+r)^t} = D_0 \left[\frac{(1+g)}{(1+r)} + \frac{(1+g)^2}{(1+r)^2} + \ldots + \frac{(1+g)^\infty}{(1+r)^\infty} \right] \qquad 8$$

If required return r is assumed to be strictly greater than growth rate g, then the square-bracketed term in Equation 8 is an infinite geometric series and sums to $[(1 + g)/(r - g)]$. Substituting into Equation 8 produces the Gordon growth model as presented in Equation 9:

$$V_0 = \frac{D_0\,(1+g)}{r-g} = \frac{D_1}{r-g} \qquad 9$$

For an illustration of the expression, suppose the current (most recent) annual dividend on a share is €5.00 and dividends are expected to grow at 4 percent per year. The required rate of return on equity is 8 percent. The Gordon growth model estimate of intrinsic value is, therefore, €5.00(1.04)/(0.08 – 0.04) = €5.20/0.04 = €130 per share. Note that the numerator is D_1 not D_0. (Using the wrong numerator is a common error.)

The Gordon growth model estimates intrinsic value as the present value of a growing perpetuity. If the growth rate, g, is assumed to be zero, Equation 8 reduces to the expression for the present value of a perpetuity, given earlier as Equation 6.

In estimating a long-term growth rate, analysts use a variety of methods, including assessing the growth in dividends or earnings over time, using the industry median growth rate, and using the relationship shown in Equation 10 to estimate the sustainable growth rate:

$$g = b \times \text{ROE} \qquad 10$$

where

g = dividend growth rate

Example 7 illustrates the application of the Gordon growth model to the shares of a large industrial manufacturing company. The analyst believes it will continue to grow at a rate that it achieved in the previous three years and remain stable in the future. The example asks how much the dividend growth assumption adds to the intrinsic value estimate. The question is relevant to valuation because if the amount is high on a percentage basis, a large part of the value of the share depends on the realization of the growth estimate. One can answer the question by subtracting from the intrinsic value estimate determined by Equation 9 the value determined by Equation 6, which assumes no dividend growth.[3]

EXAMPLE 7

Applying the Gordon Growth Model

Siemens AG operates in the capital goods and technology space. It is involved in the engineering, manufacturing, automation, power, and transportation sectors. It operates globally and is one of the largest companies in the sectors in which it operates. It is a substantial employer in both its original, domestic German market, as well as dozens of countries around the world. Selected financial information for Siemens appears in Exhibit 3.

Exhibit 3:

Selected Financial Information for Siemens AG

Year	2017	2016	2015	2014	2013
EPS	€7.45	€6.74	€8.85	€6.37	€5.08
DPS	€3.7	€3.6	€3.5	€3.3	€3.0
Payout ratio	50%	53%	40%	52%	59%
ROE	15.6%	15.9%	22.3%	18.2%	14.6%
Share price (XETRA - Frankfurt)	€119.2	€104.2	€79.94	€94.37	€89.06

Note: DPS stands for "dividends per share."
Source: Morningstar, www.siemens.com.

The analyst estimates the growth rate to be approximately 5.4 percent based on the dividend growth rate over the period 2013 to 2017 [$3(1 + g)^4 = 3.7$, so $g = 5.4\%$]. To verify that the estimated growth rate of 5.4 percent is feasible in the future, the analyst also uses the average of Siemens's retention rate and ROE for the previous five years ($g \approx 0.49 \times 17.3\% \approx 8.5\%$) to estimate the sustainable growth rate.

Using a number of approaches, including adding a risk premium to a long-term German government bond and using the CAPM, the analyst estimates a required return of 7.5 percent. The most recent dividend of €3.70 is used for D_0.

1. Use the Gordon growth model to estimate Siemens's intrinsic value.

Hide Solution

Solution to 1:

$$V_0 = \frac{€3.70\,(1 + 0.054)}{0.075 - 0.054} = €185.70$$

2. How much does the dividend growth assumption add to the intrinsic value estimate?

Hide Solution

Solution to 2:

overvalued, or fairly valued?

Hide Solution

Solution to 3:

A share of Siemens appears to be undervalued. The analyst, before making a recommendation, might consider how realistic the estimated inputs are and check the sensitivity of the estimated value to changes in the inputs.

4. What is the intrinsic value if the growth rate estimate is lowered to 4.4 percent?

Hide Solution

Solution to 4:

$$V_0 = \frac{€3.70\,(1 + 0.044)}{0.075 - 0.044} = €124.61$$

5. What is the intrinsic value if the growth rate estimate is lowered to 4.4 percent and the required rate of return estimate is increased to 8.5 percent?

Hide Solution

Solution to 5:

$$V_0 = \frac{€3.70\,(1 + 0.044)}{0.085 - 0.044} = €94.21$$

The Gordon growth model estimate of intrinsic value is extremely sensitive to the choice of required rate of return *r* and growth rate *g*. It is possible that the growth rate assumption and the required return assumption used initially were too high. Worldwide economic growth is typically in the low single digits, which may mean that a large company such as Siemens may struggle to grow dividends at 5.4 percent into perpetuity. Exhibit 4 presents a further sensitivity analysis of Siemens's intrinsic value to the required return and growth estimates.

Exhibit 4:

Sensitivity Analysis of the Intrinsic-Value Estimate for Siemens AG

	g = 2.5%	*g* = 3.5%	*g* = 4.5%	*g* = 5.5%	*g* = 6.5%
r = 6%	€108.4	€153.2	€257.8	€780.7	—
r = 7%	€84.3	€109.4	€154.7	€260.2	€788.1
r = 8%	€69.0	€85.1	€110.5	€156.1	€262.7
r = 9%	€58.3	€69.6	€85.9	€111.5	€157.6
r = 10%	€50.6	€58.9	€70.3	€86.7	€112.6

Note that no value is shown when the growth rate exceeds the required rate of return. The Gordon growth model assumes that the growth rate cannot be greater than the required rate of return.

The assumptions of the Gordon model are as follows:

- Dividends are the correct metric to use for valuation purposes.
- The dividend growth rate is forever: It is perpetual and never changes.
- The required rate of return is also constant over time.

simplistic to reflect the characteristics of the companies being evaluated. Some alternatives to using the Gordon model are as follows:

- Use a more robust DDM that allows for varying patterns of growth.
- Use a cash flow measure other than dividends for valuation purposes.
- Use some other approach (such as a multiplier method) to valuation.

Applying a DDM is difficult if the company being analyzed is not currently paying a dividend. A company may not be paying a dividend if 1) the investment opportunities the company has are all so attractive that the retention and reinvestment of funds is preferable, from a return perspective, to the distribution of a dividend to shareholders or 2) the company is in such shaky financial condition that it cannot afford to pay a dividend. An analyst might still use a DDM to value such companies by assuming that dividends will begin at some future point in time. The analyst might further assume that constant growth occurs after that date and use the Gordon growth model for valuation. Extrapolating from no current dividend, however, generally yields highly uncertain forecasts. Analysts typically choose to use one or more of the alternatives instead of or as a supplement to the Gordon growth model.

EXAMPLE 8

Gordon Growth Model in the Case of No Current Dividend

1. A company does not currently pay a dividend but is expected to begin to do so in five years (at $t = 5$). The first dividend is expected to be $4.00 and to be received five years from today. That dividend is expected to grow at 6 percent into perpetuity. The required return is 10 percent. What is the estimated current intrinsic value?

Hide Solution

Solution:

The analyst can value the share in two pieces:

1. The analyst uses the Gordon growth model to estimate the value at $t = 5$; in the model, the year-ahead dividend is $4(1.06). Then the analyst finds the present value of this value as of $t = 0$.
2. The analyst finds the present value of the $4 dividend not "counted" in the estimate in Piece 1 (which values dividends from $t = 6$ onward). Note that the statement of the problem implies that D_0, D_1, D_2, D_3, and D_4 are zero.

Piece 1: The value of this piece is $65.818:

$$V_n = \frac{D_n(1+g)}{r-g} = \frac{D_{n+1}}{r-g}$$

$$V_5 = \frac{\$4(1+0.06)}{0.10-0.06} = \frac{\$4.24}{0.04} = \$106$$

$$V_0 = \frac{\$106}{(1+0.10)^5} = \$65.818$$

Piece 2: The value of this piece is $2.484:

$$V_0 = \frac{\$4}{(1+0.10)^5} = \$2.484$$

The sum of the two pieces is $65.818 + $2.484 = $68.30.

Alternatively, the analyst could value the share at $t = 4$, the point at which dividends are expected to be paid in the following year and from which point they are expected to grow at a constant rate.

The next section addresses the application of the DDM with more flexible assumptions as to the dividend growth rate.

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Discussion

What do you want to discuss... (required)

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Example 8

Why is there a Piece 2? The Gordon

Growth Model takes that piece into account by taking D5 and multiplying it by the growth factor making it into D6?

I.e.: V5 = D6 / (r – g) = ($4 * 1.06) / (0.1 – 0.06) = $106

Why is there a need to separately discount the first dividend if it's already included in V5 (which is based on D6 onward)?

The $106, isn't this the entire intrinsic value?

If we include Piece 2, where we discount the $4 to the PV, aren't we double counting? Gordon Growth Model assumes all dividends from t = 6 onward, and the $4 at t = 5 is not part of that perpetuity.

If the question intended to include the $4 at t = 5 as an extra dividend, it would have said so explicitly (e.g., "$4 at t = 5, then growth starts at t = 6"). But here, the $4 is the first dividend, and growth applies immediately after. So, we take that $4, take it to next year, i.e., D6, and apply the Gordon Growth Model? Just like we would if the dividends were to start immediately, we don't take the dividend in year 1 and add it to the Gordon Growth Model.

Created 4 months ago by Maxim Ovchinnikov

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